SIMATS ENGINEERING
Department of Computer Science and Engineering

ASSESSMENT TOOL2 -CONCEPT MAPPING

Course Code:	CSA12	**Course Title:**	Computer Architecture
CO Assessed:	**CO1**	**Assessment:**	**ASSESSMENT TOOL 2-** CONCEPT MAPPING
Weightage:	**35%**	**Total Marks:**	**25**
CO1Statement:	Analyze the functional units of a computer system including CPU, memory, bus structures, and I/O components by examining instruction execution cycles, addressing modes, and performance metrics such as CPI, clock cycles, and benchmarking(BL4)		
Student Name:		**Reg.No.:**	
Department:		**Date:**	

ANNEXURE A

1. Construct a detailed concept map showing the relationship between CPU (ALU, Control Unit, Registers), memory (cache, RAM), and I/O devices. Extend the map to include single-bus, multi-bus, and hierarchical bus structures, and clearly illustrate how data and control signals flow among these components during execution. Indicate possible points of delay or bottlenecks and how different bus structures help in improving performance.
2. Develop a comprehensive concept map linking the instruction execution cycle (fetch, decode, execute) with CPU performance parameters such as CPI, clock cycles, and execution time. Show how each stage contributes to overall execution time, and include connections to factors like memory access, instruction complexity, and pipeline stalls. Also, represent how benchmarking is used to evaluate performance.
3. Create a concept map comparing 8085 and 8086 architectures, including their instruction sets, register organization, addressing modes, and memory segmentation (in 8086). Clearly show how these architectural features influence program execution, multitasking capability, and efficiency in real-time applications.
4. Draw a detailed concept map that connects various addressing modes (immediate, direct, register, indirect, indexed) with instruction types (data transfer, arithmetic, control instructions). Illustrate how each addressing mode affects memory access time, instruction size, and execution speed, and show real-world scenarios where specific addressing modes are preferred.
5. Prepare an elaborate concept map integrating number systems (binary and hexadecimal) with instruction representation, memory storage, and debugging processes. Show how data is converted between formats, how instructions are stored and interpreted by the CPU, and why hexadecimal representation is commonly used in low-level programming and system design.

Code of Conduct:

I Jeevan Raaja S (129421005) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Jeevan Raaja S

MARKS ALLOCATION

Criteria	Concept Identification (25%)	Linkages (25%)	Organization (20%)	Integration of Literature (20%)	Clarity (10%)
Q1					
Q2					
Q3					
Q4					
Q5					

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts

Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

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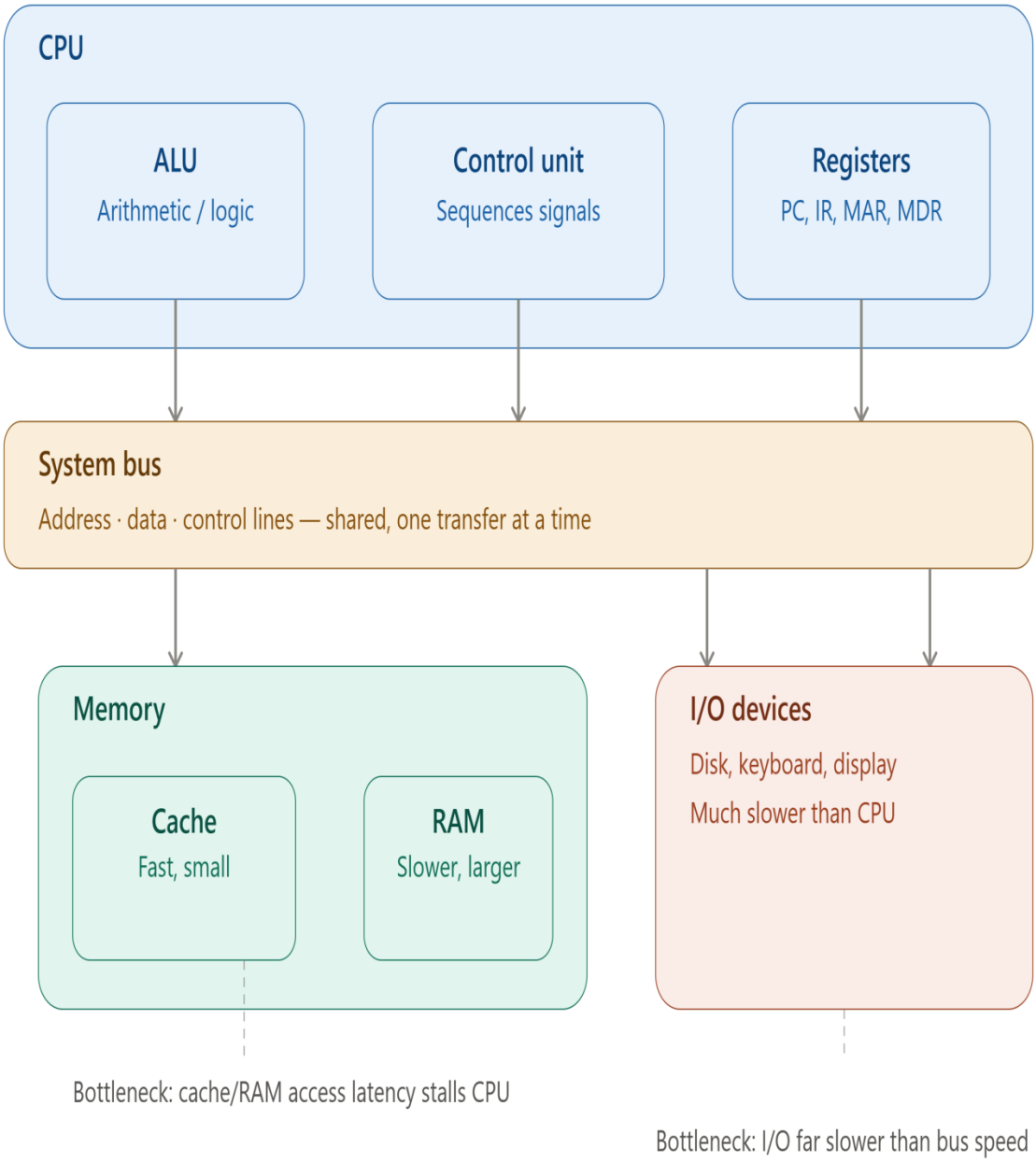
CPU
ALU
Arithmetic / logic
Control unit
Sequences signals
Registers
PC, IR, MAR, MDR
System bus
Address · data · control lines — shared, one transfer at a time
Memory
Cache
Fast, small
RAM
Slower, larger
I/O devices
Disk, keyboard, display
Much slower than CPU
Bottleneck: cache/RAM access latency stalls CPU
Bottleneck: I/O far slower than bus speed

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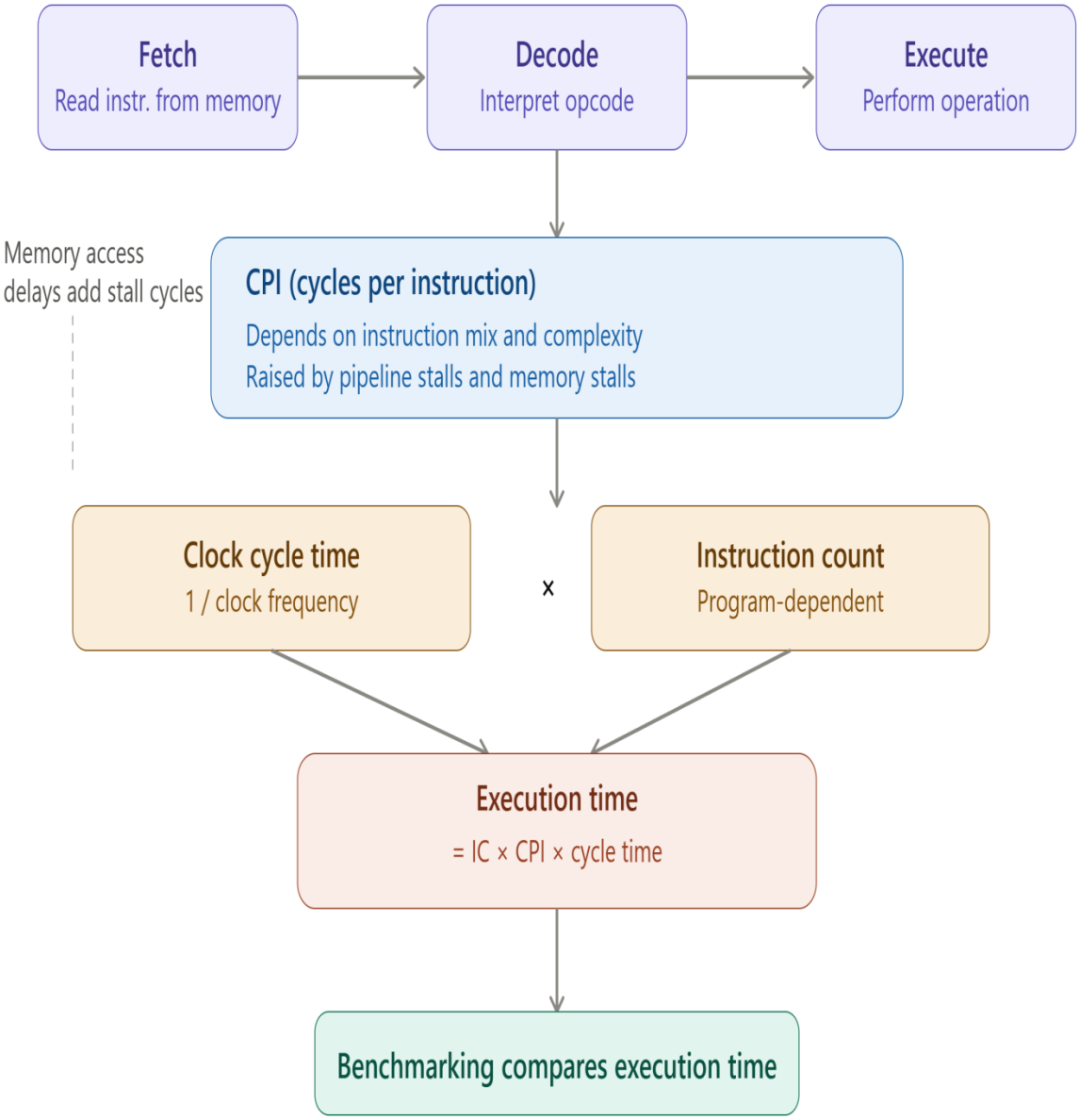

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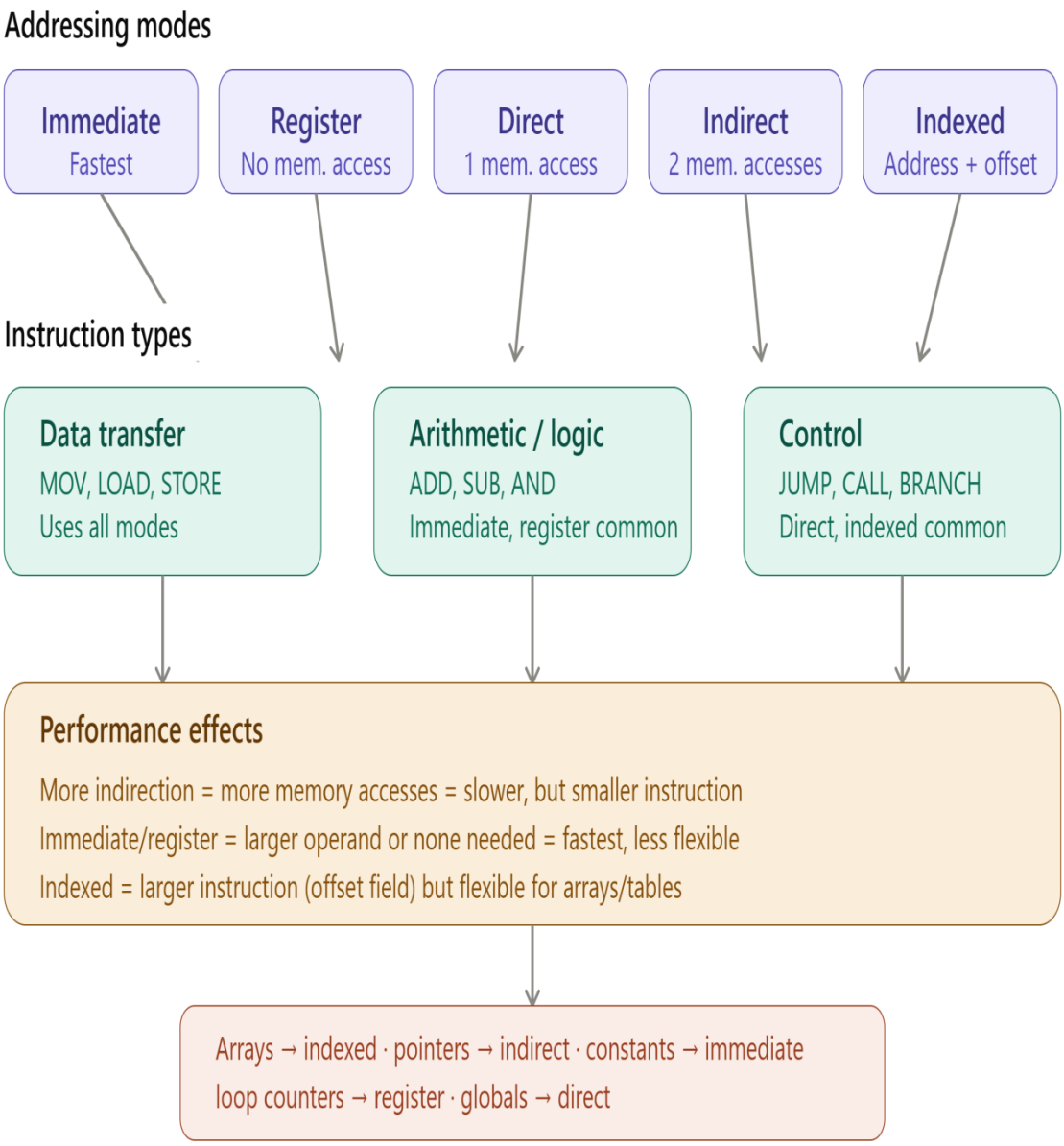

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